

] What is ANN?

An ANN (Artificial Neural Network) is a machine learning model inspired by the human brain, consisting of neurons organized in layers:

- Input layer
- Hidden layer(s)
- Output layer

How it works?

- ① Input features are fed into the network
- ② Each neuron computes a weighted sum
- ③ Activation function is applied
- ④ Output layer is passed to next layer
- ⑤ Final prediction is produced

Mathematical Implementation:

Neuron computation:

$$z = \sum (w_i x_i) + b$$

$$a = f(z)$$

where x_i - input

w_i - weight

b - bias

f - activation function

Common Activation Functions:

• ReLU : $f(x) = \max(0, x)$

• Sigmoid : $f(x) = \frac{1}{1 + e^{-x}}$

Loss function:

$$L = \frac{1}{n} \sum (y - \hat{y})^2$$

Weight updated (Gradient Descent):

$$w = w - \eta \frac{\partial L}{\partial w}$$

2] What is CNN?

A CNN (Convolutional Neural Network) is specialized for image processing & uses convolution operation to extract features.

How it works?

- ① Convolution layer extracts features
- ② Activation (ReLU)
- ③ Pooling reduces dimensions
- ④ Fully connected layer for classification

Mathematical Implementation:

Convolution:

$$(S * K)(i, j) = \sum_m \sum_n S(i+m, j+n) \cdot K(m, n)$$

where:

- S - input image
- K - kernel / filter

Pooling (Max Pool):

$$y = \max(x_{\text{region}})$$

3] Recurrent Neural Network (RNN)

An RNN is used for sequential data, (text, timeseries) & has memory of previous inputs.

How it works?

- Processes one element at a time
- Uses previous hidden state
- Maintains sequence context

Mathematical implementation:

$$h_t = \tanh (w_h h_{t-1} + w_x x_t + b)$$

$$y_t = w_y h_t$$

A] LSTM (Long Short-Term Memory)

An advanced RNN that solves vanishing gradient problem using gates.

How it works?

Uses 3 gates:

- Forget gate
- Input gate
- Output gate

Mathematical Implementation:

Forget gate: $f_t = \sigma(w_f [h_{t-1}, x_t])$

Input gate: $i_t = \sigma(w_i [h_{t-1}, x_t])$

$$\tilde{C}_t = \tanh(w_c [h_{t-1}, x_t])$$

Cell state: $C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$

Output gate:

$$o_t = \sigma(w_o [h_{t-1}, x_t])$$

$$h_t = o_t * \tanh(C_t)$$

5] What is GRU?

A Gated Recurrent Unit (GRU) is a simplified version of LSTM with fewer gates.

How it works?

It uses:-

- update gate
- reset gate

Mathematical implementation:

Update gate:

$$z_t = \sigma(w_z [h_{t-1}, x_t])$$

Reset gate:

$$r_t = \sigma(w_r [h_{t-1}, x_t])$$

Hidden state:

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tanh(w_h [r_t * h_{t-1}, x_t])$$

6] What is GAN?

A Generative Adversarial Network (GAN) consists of two networks:

- Generator (creates fake data)
- Discriminator (detects real vs fake)

How it works?

- ① Generator produces fake samples
- ② Discriminator evaluates them
- ③ Both networks compete

Mathematical Implementation:

Objective function:

$$\min_G \max_D (D, G)$$

$$V(D, G) = \mathbb{E} [\log D(x)] + \mathbb{E} [\log (1 - D(G(z)))]$$

7] What is VAE?

A VAE (Variational AutoEncoder) is a generative model that learns probability distributions of data.

How it works?

- ① Encoder maps input $\rightarrow$ latent space
- ② Latent space sampling
- ③ Decoder reconstructs data

Mathematical Implementation:

Latent Distribution:

$$z \sim \mathcal{N}(\mu, \sigma^2)$$

Loss function:

$L = \text{Reconstruction Loss} + \text{KL Divergence}$

$$\text{KL} (q(z|x) \parallel p(z))$$